

Skill & Product Library / Sales Automation

Claude Skill

Shipped

Clay Prospecting Engine

Zero-UI prospecting for non-technical reps — upload a CSV, get an enriched list back.



Two-mode prospecting, zero Clay UI interaction required from reps

Claude Skill

Clay

HubSpot

enrichment

prospecting

Overview

Clay is a powerful GTM enrichment platform, but reps at a global HR-tech (EOR) platform couldn't (and shouldn't have to) drive its UI — every list request bottlenecked on one person. This production Claude skill lets a rep upload any CSV; it auto-detects whether it's a company list or a contact list, routes to the right Clay pipeline via the Clay REST API, and returns enriched results — synced to a CRM table, dropped as a CSV in chat, or emailed. Two-mode prospecting (Company → find & enrich contacts; Contacts → fill missing emails/phones/LinkedIn/firmographics) with zero Clay UI interaction required. Backed by a full AI Revenue Operations agent charter governing source-of-truth boundaries, an 18-code suppression taxonomy, and a safe CRM-update policy.

The Problem

Clay is genuinely powerful for enrichment, but its table-building UI isn't something 50+ non-technical reps can (or should) drive themselves. Every list request — "enrich this list of companies," "fill in missing emails for these contacts" — routed through one person, creating a queue and a single point of failure.

How It Works

1. **Upload any CSV** — a rep drops a spreadsheet into the Claude skill; no Clay account or login needed on the rep's side.
2. **Header-signature mode detection** — the skill inspects the column headers to auto-detect whether the upload is company-level or contact-level data, and picks the matching Clay pipeline without the rep specifying anything.
3. **Company mode** — for a company list, the skill routes to a Clay table via the **Clay REST API** (Clay's programmatic interface, letting other tools trigger its enrichment without opening Clay itself) that finds and enriches decision-maker contacts at each company.
4. **Contact mode** — for a contact list, a different Clay pipeline fills missing emails, phones, LinkedIn URLs, and firmographic data on existing records.
5. **Staged execution protocol** — the skill runs in defined stages (validate → submit to Clay → poll for completion → format results) rather than a single black-box call, so failures are catchable and reportable mid-run.
6. **Delivery choice** — results return as a **CRM table sync** (writes directly into HubSpot), a CSV attached in the chat, or an emailed file — whichever fits the rep's workflow.
7. **Governance underneath** — every run operates inside an AI Revenue Operations charter: the CRM is the source of truth, Clay is enrichment-only, a workflow-automation layer handles orchestration, and a backfill data vendor covers gaps; an 18-code suppression taxonomy (customers, open deals, active sequences, bounces, DNC, etc.) prevents prospecting into people who shouldn't be prospected; CRM writes fill blanks but never blind-overwrite existing data, flagging conflicts instead.



Stack & Integrations

Tool	Role	Access needed
Claude skill	CSV ingestion, mode detection, staged execution, delivery	Claude Code / Claude with skill installed
Clay REST API	Programmatic trigger of company/contact enrichment pipelines	Clay API key + pre-built table IDs
CRM (HubSpot)	System of truth; optional direct table sync destination	Private-app token (contacts/companies read-write)
CSV parsing/validation	Header-signature detection, input validation	Ships with the skill

Setup Process

1. Build (or reuse) the two Clay pipelines the skill routes to: one for company→contact enrichment, one for contact backfill.
2. Install the Claude skill; configure the Clay API key and the two table IDs it targets.
3. Configure the CRM private-app token if direct table sync is wanted; otherwise chat/email delivery works standalone.
4. Test with a small sample company list and a small sample contact list to confirm mode auto-detection picks correctly.
5. Set suppression rules (18-code taxonomy) matching your CRM's own do-not-contact / customer / active-sequence fields.
6. Roll out to the rep team with a short usage guide; monitor the first week's runs against the governance charter.

What You Need

- [] Clay workspace with the two target pipelines built and an API key
- [] Claude Code or Claude access for the team
- [] Optional: CRM private-app token for direct table sync
- [] A defined suppression policy (customers, open deals, DNC, bounces, active sequences, etc.)
- [] ~1 day to configure pipelines + test both modes

Results

- **Shipped** with a rep-facing guide.
- Two-mode prospecting (Company and Contacts) with **zero Clay UI interaction** required from reps.
- Backed by a full agent charter: source-of-truth boundaries per system, safe CRM-update policy (fill blanks, never blind-overwrite, flag conflicts), audit trail per run, and a secrets policy that forbids pasted credentials.

Challenges & Fixes

- **Every enrichment request bottlenecked on one person** → self-serve skill with auto-detected mode, removing the manual triage step entirely.
- **Risk of reps overwriting good CRM data with enrichment results** → safe-update policy: fill blanks only, never blind-overwrite, flag conflicts for review instead of silently resolving them.
- **Risk of re-prospecting suppressed contacts (customers, DNC, active sequences)** → an 18-code suppression taxonomy enforced before any list goes out.
- **Credential handling by non-technical users** → a secrets policy that forbids pasting credentials into chat; the skill holds the Clay API key, not the rep.



Demonstrates

Productizing internal tooling for non-technical users · compliance-first outbound design · agent governance documents that read like platform policy, not a README.

STACK & INTEGRATIONS

Claude Skill

Clay

HubSpot

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DISCUSS THIS BUILD

Happy to walk through the architecture, the failures along the way, or how this could translate to your team.

Email Patrick

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